

The Data End of Crypto Quant for Public LLMs: An Observation Layer that Compiles World Bundles

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Abstract

Purpose. End-to-end crypto quant for public LLMs needs a *data end* before decision or execution. We build that layer as an anonymized observation runtime: multi-band evidence $\rightarrow$ BandPIT/WMI $\rightarrow$ a point-in-time (PIT) *world bundle*, with hard refusal when support is thin. **Evidence.** Primary—seeing: grounding recovers ready/missing/tilt at ≈ 0.97 under Compiled vs. $\lesssim 0.23$ under Raw, while sufficiency chat alone is unreliable (Compiled mean 0.60). Secondary—valve: Compiled thin-abstain is 1.0 (100-day and full OOS $n=2000$) while Ungated soft judgment is only $\approx 0.68 / \approx 0.56$; Raw over-trades. Secondary—content: vintaged tilts beat momentum ($\Delta CE=0.334$, $p=0.034$; LOBO content share 1.0); open-slice Ungated CE is point-positive as a field lower bound. The panel is scarce (max WMI $\approx 0.093 < 0.2$; five schema bands permanently missing), so production Compiled never opens—a valve stress test. **Claim.** Build the quant data end first: compile so models can see; refuse when they must not act.

CCS Concepts

• **Computing methodologies** $\rightarrow$ **Artificial intelligence**; *Machine learning*; • **Applied computing** $\rightarrow$ *Economics*.

Keywords

data end, observation layer, world bundle, grounding, public LLMs, cryptocurrency, point-in-time, quality-gated refusal, systems

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1 Introduction

Purpose (read this first). Build the *data end* of end-to-end crypto quant for public LLMs. The stack is **data/observation** $\rightarrow$ **decision** $\rightarrow$ **execution** (Table 1); agents and generative world models [6, 7, 9, 19–21] typically assume the first stage. Crypto violates that assumption: exchange, macro, and alternative evidence arrive on incompatible clocks [11, 13]. We own only the data end—compile a PIT world bundle so models can *see*, and hard-refuse when the view is too thin. Refusal is a safety valve, not the product and not a trading claim.

Failure modes. No feed $\rightarrow$ refuse (safe, useless as a data end). Thin price fragment $\rightarrow$ trade and lose (unsafe for any downstream stack). Disclosed but ungated world $\rightarrow$ read fields, disagree on action. So the data end must (i) make the market legible and (ii) enforce a quality valve before decision/execution.

Table 1: End-to-end quant stack; this paper owns only the data end.

Stage	Typical owner	This paper
Data / observation	Assumed by agents	World bundle + quality valve
Decision	LLM / policy / portfolio	Out of scope
Execution	Broker / OMS	Out of scope

What we build. An anonymized runtime (Section 4): multi-band evidence $\rightarrow$ BandPIT/WMI $\rightarrow$ JSON world bundle for GPT / DeepSeek / GLM / Gemini class models, served through a read API a decision layer can call. “World model” means state compiler + quality gate (Section 3), not dynamics and not a decision engine. The bundle is complete, honest, and auditable; `should_ai_abstain` is the machine-readable valve.

How we evaluate. **RQ1 (primary—seeing):** Are ready / missing / tilt answers verifiable against PIT ground truth under Compiled vs. Raw? **RQ2 (valve):** Does a hard flag enforce thin abstention where soft disclosure does not? Four arms: Compiled, Ungated (same content, no flag), Raw (mom5), Blind. **RQ3 (secondary):** Are vintaged tilts nonempty under a transparent no-LLM rule?

Contributions. (1) Data-end semantics: epistemic observations, lag reconstruction, Π_t , and seeing vs. gating. (2) Anonymized PIT compile path: layered collectors $\rightarrow$ BandPIT $\rightarrow$ service surface $\rightarrow$ bundles; schema honesty (eight bands, three live); frozen four-arm harness. (3) Live + rule validation: grounding ≈ 0.97 with sufficiency disassociation; thin-abstain 1.0 at 100-day and full-OOS scale; nonempty tilts ($\Delta CE=0.334$; LOBO content share 1.0) and an open-slice lower bound. Estimand: legibility and valve reliability—not decision SOTA, execution, or Compiled-open alpha.

2 Related Work

World models. World models learn compact states and often dynamics [6, 7, 9]. We address the complementary observation-side problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy—not a Dreamer/JEPA simulator.

LLM agents in finance. Return prediction [12], open financial LLMs [18], memory- and tool-augmented agents [10, 19–21], and ICAIF retrieval/debate/judge systems [1, 3, 15, 16] improve how agents fetch, debate, and are scored. They still assume a usable observation stream at the data end of the stack. We study that prior stage: compile a world bundle, measure whether models can *see* it, then gate action when they must not act. Our estimand is therefore orthogonal to memory routers and debate judges: without a typed PIT world, those stacks consume whatever fragment the tool dump provides.

Selective prediction and PIT. Abstention can be optimal under noise [4, 5]; we attach refusal to world quality via a runtime field,

not classifier confidence alone. Macro vintages [2] and crypto microstructure [11, 13] motivate multi-band compilation with disclosed missingness. Bootstrap discipline for the content probe follows [8, 14, 17].

3 Observation Layer

Definition 3.1 (Epistemic observation). $O_{j,t} = (x, \tau, q, g, r)$: value, latest-available time, quality, main-view gate, semantic role. Omitting (τ, q, g, r) yields a feature dump, not a world observation.

Definition 3.2 (World bundle). $W_t = \Pi_t(\mathcal{F}_t^{\text{raw}})$ with scalar quality $q(W_t)$ (WMI/ACWMI); $\mathcal{F}_t^{\text{AI}} = \sigma(W_t)$. Π_t aligns clocks, gates honesty/missingness, and emits consumer JSON—the handoff to decision.

Completeness, honesty, quality. Ready/limited/missing counts disclose coverage. $\text{WMI}_t = B_t U_t H_t$ aggregates breadth, stability, and honesty; ACWMI adds regime weight. Compiled exposes `should_ai_abstain` and `thin_world`.

Seeing vs. gating. The bundle is what the model *sees*; the hard flag *blocks* action when $q(W_t)$ is too low. Tilts without a gate are a dump; a gate without a readable bundle is only a switch.

Assumption 3.3 (Bounded lag and increments). Band j has max observation lag Δ_j ; latent increments satisfy $\|S_u - S_{u-1}\| \leq \bar{\delta}$.

PROPOSITION 3.4 (LAG RECONSTRUCTION BOUND). *Under Assumption 1, $\|\tilde{S}_t - S_t\| \leq C_{\text{del}}\bar{\delta} + C_{\text{noi}}\varepsilon_t + C_{\text{miss}}m_t$. Compilation cannot erase delay; it can refuse to pretend the bound is zero.*

Intuition. Agents that concatenate tool dumps silently absorb delay and missingness into the feature vector. Prop. 3.4 makes those terms explicit: a thicker raw payload can worsen reconstruction if honesty collapses. The data end therefore grades readiness and discloses gaps instead of imputing a false full world.

PROPOSITION 3.5 (COMPILATION $\neq$ DUMP). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\text{AI}}$: ungated stale evidence can raise volume while lowering honesty.*

PROPOSITION 3.6 (WORLD-CONDITIONAL ABSTENTION). *If non-abstain actions exceed a world-dependent cost, the Bayes action is abstain; with costs decreasing in WMI, abstention forms a lower set in world quality [4].*

PROPOSITION 3.7 (LOBO CONTENT VS. GATING). *Deleting band j changes value through a content channel and a gating channel. Empirically we report content share under an ungated rule so the probe speaks to nonempty fields.*

Implication for the runtime. Props. 3.5–3.6 say the data end must export gates and abstain flags, not tilts alone: compilation without quality disclosure is still a feature dump; disclosure without a hard gate still lets thin worlds drive action. Prop. 3.7 motivates the economic check in RQ3: if deleting a durable band does not change the content rule, the bundle is empty of content.

REMARK 3.8 (NOT A GENERATIVE WM). *We do not learn $p(s_{t+1} | s_t, a_t)$. We deliver a state compiler with a quality-gated refuse interface—the data end of the stack.*

Table 2: Compile-path modules (systems view, not a product inventory).

Module	Role
Collectors / stores	Timestamped / vintaged band history
BandPIT compiler	Previous-close readiness; Π_t
Quality indices	$B, U, H \rightarrow$ WMI; abstain flag; O_t
Bundle builder	Complete / honest / auditable JSON
Service surface	Bundle / health / availability reads
Consumer harness	Four-arm protocol; temp. 0

4 Runtime: Data-End Compile Path

The anonymized prototype implements the data end as a layered compile path (Figure 1, Table 2). Decision and execution sit downstream; this paper does not claim them. Paths and product names are omitted for double-blind review.

4.1 Layered design

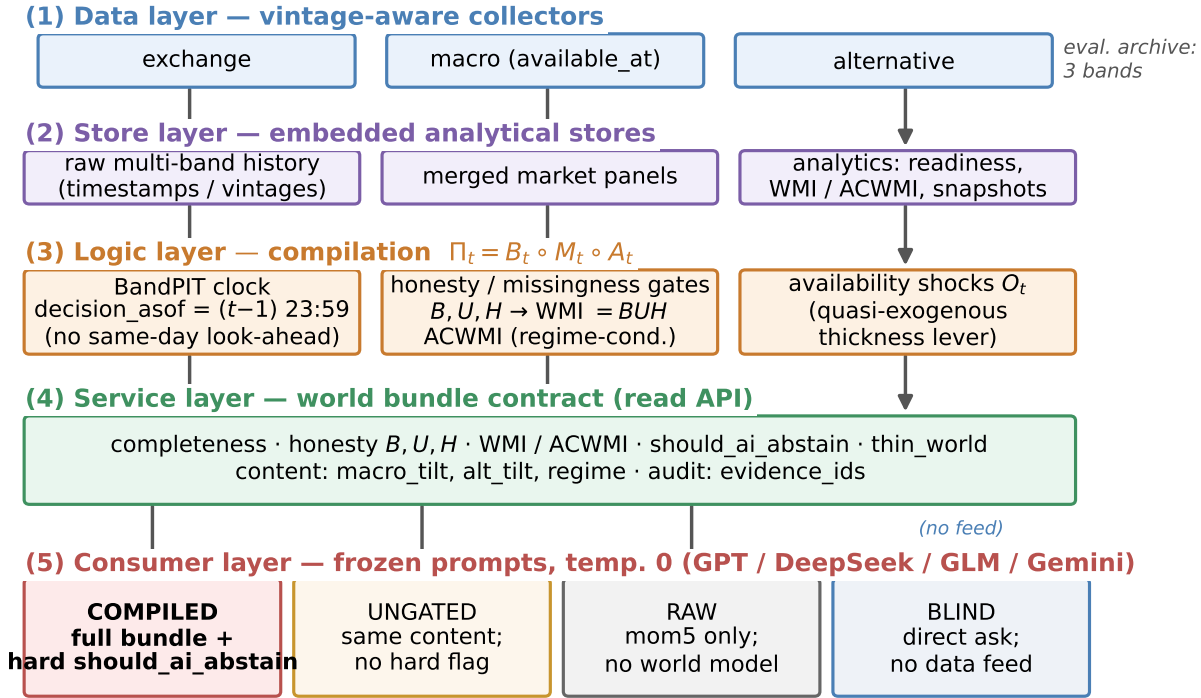
- (1) **Data layer.** Exchange, macro, and alternative collectors write vintage-aware history (observation timestamps; macro `available_at`).
- (2) **Store layer.** Embedded stores hold raw series, merged market panels, and compiled analytics (readiness, WMI/ACWMI, world snapshots).
- (3) **Logic layer.** BandPIT reconstructs readiness on a previous-close clock; honesty/missingness gates implement Π_t ; WMI/ACWMI and availability shocks O_t (band outage events for quasi-exogenous levers) are first-class.
- (4) **Service layer.** A read API serves quality-tagged bundles and health objects without restarting collectors for newly arrived rows. Anonymized surface includes world-bundle and decision-bundle reads, health exposing `should_ai_abstain`, a paper world-model time slice, and availability-shock queries—the machine-readable handoff a decision layer would call.
- (5) **Consumer layer.** Frozen-prompt LLM/rule agents call an OpenAI-compatible adapter at temperature 0 under four arms.

4.2 Archive schema vs. live evaluation bands

The runtime schema is band-extensible (exchange, macro, alternative, news, on-chain, options, tokenomics, event calendar). On the evaluation panel (≈ 3990 asset-days; 2025-07 to 2026-08), only three bands carry history: exchange ready share ≈ 0.80 , macro 1.0, alternative 1.0; the other five are permanently missing. That structural hole caps breadth and keeps max WMI ≈ 0.093 even on 3/3-ready days—honesty about what the data end can currently show, not a claim that denser bands never help. Table 3 lists the three live bands that populate the consumer object.

4.3 PIT clock and freshness

For calendar day t , BandPIT reconstructs the decision information set at $(t-1)$ 23:59 (payoff is same-day close-to-close return); the



Live abstention outcomes: Sec.~Experiments (Table / four-arm figure).

Figure 1: Data-end pipeline: evidence → BandPIT → world bundle → Compiled / Ungated / Raw / Blind.

Table 3: Input evidence vs. fields emitted in the world bundle (live archive).

Band	Fetches (raw)	Emitted (bundle)
exchange	OHLCV, funding, OI	mom5; status/age
macro	equity-vol / dollar indices	macro_tilt; status/age
alternative	stablecoin circulation	alt_tilt; status/age
derived	—	WMI, ACWMI, abstain, O_t , evidence IDs

bundle records decision_asof. Each band is graded ready /limited /missing against daily-scale age thresholds (Table 4). Missingness is disclosed rather than imputed (Prop. 3.4). Breadth, stability, and honesty aggregate into $WMI_t = B_t U_t H_t$ (ACWMI adds regime weight). PIT-safe tilts use only in-clock observations: five-day macro changes, a seven-day stablecoin-flow slope, and mom5. The valve sets should_ai_abstain when $WMI < 0.2$ and exposes thin_world. Figure 2: macro and alternative stay ready; exchange readiness varies and defines thick/thin days. Figure 3: WMI stays below the production gate throughout; ACWMI can look healthier under regime weight, so the runtime exports both and lets the configured index mode drive abstain. Table 5: production gate 0.2 never opens on this panel; $WMI \geq 0.05$ selects the band-thick subset ($\approx 61\%$ OOS).

Table 4: BandPIT freshness thresholds (seconds → days).

Band	Fresh window	In eval archive
exchange	2d	yes (ready ≈ 0.80)
macro	5d	yes (ready 1.0)
alternative	7d	yes (ready 1.0)
news / on-chain / options	3d	missing
tokenomics	7d	missing
event calendar	14d	missing

Table 5: WMI gate vs. open share (OOS). Production never opens on this panel.

Gate	Open share	Note
0.2 (production)	0.00	max WMI ≈ 0.093
0.05 (counterfactual)	0.61	$\equiv$ band-thick
0.01	1.00	near-floor open

4.4 World bundle (what the LLM sees)

The consumer object is a JSON world bundle (Table 6): timing, completeness, honesty, quality (including the abstain flag), content tilts, and evidence_ids. Listing 1 shows an exchange-gap day; Listing 2 shows a 3/3-ready day that still carries $WMI \approx 0.093$

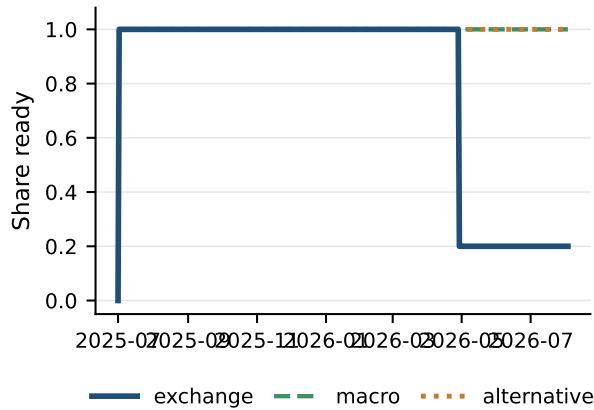


Figure 2: PIT readiness: what is visible in the archive over time.

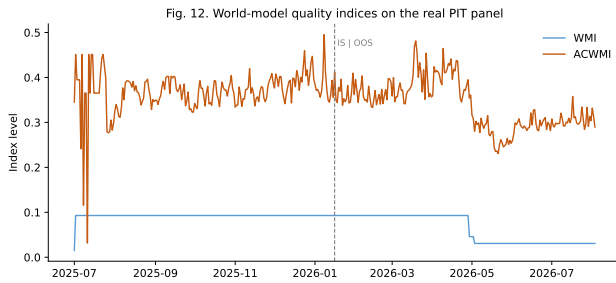


Figure 3: World-quality paths: WMI stays under the 0.2 production gate; ACWMI is regime-weighted.

Table 6: World-bundle field groups (the market view).

Group	Fields
Timing	decision_asof, previous-close protocol
Completeness	ready/limited/missing counts, band status
Honesty	B, U, H; main-view / stale gates
Quality	WMI, ACWMI, abstain flag, thin_world
Content	macro_tilt, alt_tilt, mom5
Audit	evidence_ids; EAR required

and `should_ai_abstain=true` because five schema bands remain missing—the data end discloses that ceiling rather than inventing coverage. Paper-lab smoke on the production engines confirms WMI/ACWMI and abstain wiring under the configured thresholds.

Listing 1: Exchange-gap bundle (Compiled).

```
{
  "date": "2026-05-11", "asset": "ADA",
  "decision_asof": "2026-05-10T23:59:00",
  "completeness": {"n_ready": 2, "n_missing": 1},
  "world_model_index": {"wmi": 0.015,
    "should_ai_abstain": true, "thin_world": true},
  "macro_tilt": 1.0, "alt_tilt": -1.0,
  "audit": {"evidence_ids": [
    "band:exchange:missing", "band:macro:ready"]}
}
```

Listing 2: 3/3-ready day still thin under schema ceiling.

```
{
  "date": "2026-01-16", "asset": "ADA",
  "completeness": {"n_ready": 3, "n_missing": 0},
  "honesty": {"H": 0.69, "U": 0.38, "B_hier": 0.35},
  "world_model_index": {"wmi": 0.093, "acwmi": 0.45,
    "should_ai_abstain": true, "thin_world": true},
  "band_status": {"exchange": "ready", "macro": "ready",
    "alternative": "ready"}}
}
```

4.5 Four arms and frozen prompts

Actions $\in \{\text{bullish, bearish, neutral, abstain}\}$. **Compiled:** full bundle; frozen prompt *requires* abstain when `should_ai_abstain` is true (and when the world is judged thin). **Ungated:** same content, completeness, and numeric WMI; no hard flag; soft prompt asks the model to judge thinness. **Raw:** mom5 only—common price-only wiring for the ladder, not an exhaustive tool-agent baseline; the estimand is the four-arm ranking of the data end, not ΔCE vs. the strongest agent. **Blind:** “Today is d . How should I trade a ?” with no feed. max WMI ≈ 0.093 , so production Compiled abstains on every OOS day—a scarce-support stress test of the valve, not a claim the data end never allows action. Compiled vs. Ungated differ only in enforcement (Listings 3–4); prompts, IS/OOS cut, and seeds are frozen. On thin Compiled days the evidence-alignment proxy (EAR) is 1.0 across vendors: rationales stay bound to disclosed `evidence_ids`.

Listing 3: Compiled (excerpt): hard valve.

1. If `should_ai_abstain` is true OR the world is thin, you MUST “abstain”.
2. Prefer band content over raw momentum (macro_tilt, alt_tilt are PIT-safe).
3. Rationale must match `evidence_ids`.

Listing 4: Ungated (excerpt): soft judgment.

1. Judge whether low WMI / missingness makes action unsafe; you MAY “abstain”.
2. Prefer band content when you act. NO hard abstain boolean; NO forced policy.

Why the hard valve is not a confound. Compiled abstention is partly prompt obedience, and that is intentional: the data end’s job on thin days is to *block* action, not to hope the model discovers thinness (Prop. 3.6). Ungated measures soft judgment from the same disclosed bundle; Raw measures fragment wiring; Blind measures no feed. The gating result is that the hard flag is reliable across vendors; soft refusal is not—which is why half-baked context remains dangerous without the valve.

4.6 Systems validation (engines)

Paper-lab smoke against the production engines (same thresholds as the live service) returns a wired WMI/ACWMI sample with `should_ai_abstain` consistent with the configured index mode and abstain cutoffs (WMI gate 0.2; ACWMI gate 0.35). That check is not a substitute for the four-arm OOS evaluation below; it confirms the data-end indices the consumer bundle exposes are the same objects the runtime compiles. Availability shocks O_t are queryable on the service surface for event-study levers; this paper uses them as schema honesty, not as a trading identification claim.

Table 7: Grounding (50 days): primary seeing result. C = Compiled; R = Raw.

Model	Ready F1		Missing F1		Tilt-sign	
	C	R	C	R	C	R
gpt-5.4-mini	0.94	0.00	0.94	0.31	0.94	0.20
deepseek-v4-flash	0.98	0.06	0.98	0.20	0.98	0.18
glm-5.2	0.98	0.09	1.00	0.11	0.98	0.20
gemini-3.5-flash-lite	0.96	0.03	0.96	0.30	0.96	0.18
Mean	0.97	0.04	0.97	0.23	0.97	0.19

Table 8: Sufficiency chat vs. field grounding (50 days). Seeing = fields; chat alone misleads.

Model	Suff. C	Suff. R	Ready F1 C	Ready F1 R
gpt-5.4-mini	0.78	0.98	0.94	0.00
deepseek-v4-flash	0.78	0.90	0.98	0.06
glm-5.2	0.80	0.96	0.98	0.09
gemini-3.5-flash-lite	0.04	0.94	0.96	0.03
Mean	0.60	0.95	0.97	0.04

5 Experiments

5.1 Setup

PIT panel ~ 399 days $\times 10$ liquid names (~ 3990 asset-days; 2025-07–2026-08), chronological IS/OOS 200/200 days (cut 2026-01-16). Unless noted, live arms share the same stratified 100 OOS asset-days (seed fixed), temperature 0, and four flash/mini vendors (gpt-5.4-mini, deepseek-v4-flash, glm-5.2, gemini-3.5-flash-lite) via an OpenAI-compatible adapter. Grounding uses 50 OOS days; full-OOS abstention uses $n=2000$; the no-LLM probe uses 199 OOS days after return alignment. Primary metrics: RQ1 grounding F1/tilt accuracy (sufficiency chat secondary); RQ2 thin-world abstain rate (CE as avoided loss, not alpha); RQ3 Δ CE of the content rule vs. momentum.

5.2 RQ1 (primary): Seeing—grounding workflow

If the data end works, models must report a *checkable* market view—not invent coverage from a price fragment.

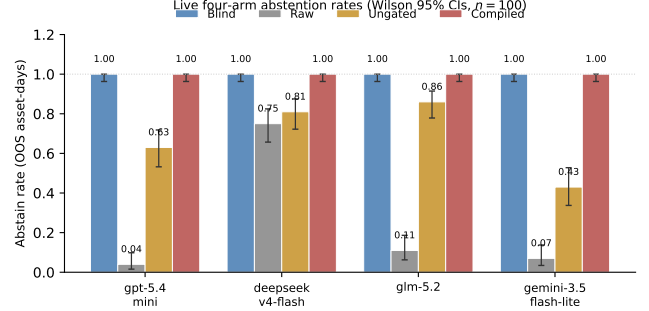
On 50 OOS asset-days, each model lists ready/missing bands among exchange/macro/alternative, reports macro_tilt / alt_tilt signs, and judges sufficiency; (i)–(iii) score against PIT ground truth.

Table 7: Compiled mean ready/missing F1 and tilt-sign accuracy 0.97; Raw collapses to 0.04, 0.23, 0.19. Raw has no readiness or tilt fields, so it cannot recover archive state; Compiled forces use of the disclosed world.

Legibility $\neq$ sufficiency chat. Table 8: Compiled sufficiency accuracy averages only 0.60 (Gemini 0.04) despite 0.97 field grounding, while Raw often answers “enough?” correctly by chance/heuristics (0.95) while inventing coverage. Chat judgment of “is this enough to trade?” is not a substitute for a typed world; we therefore measure seeing by checkable fields (RQ1) and action safety by the hard valve (RQ2).

Table 9: Thin-world abstention on identical 100 OOS days (Wilson 95% CIs). Blind = 1.0.

Model	Thin C	Thin U [CI]	Abs. R [CI]	Δ CE
gpt-5.4-mini	1.00	0.63 [.53,.72]	0.04 [.02,.10]	+0.73
deepseek-v4-flash	1.00	0.81 [.72,.87]	0.75 [.66,.82]	+1.22
glm-5.2	1.00	0.86 [.78,.91]	0.11 [.06,.19]	+1.04
gemini-3.5-flash-lite	1.00	0.43 [.34,.53]	0.07 [.03,.14]	+1.30
Mean	1.00	0.68	—	+1.07

**Figure 4: Safety valve: Blind refuses; Raw over-trades; Ungated is unstable; Compiled enforces the hard gate.**

Case study. Listings 5–6: ADA (2026-05-11) exchange missing; Compiled recovers the checkable view; Raw cannot. A second day (2026-01-17) shows the valve contrast: Compiled abstains on should_ai_abstain; Raw goes bullish from mom5 = +1 alone.

Listing 5: Compiled grounding (gpt-5.4-mini, ADA 2026-05-11).

```
ready: alternative, macro
missing: exchange
macro_tilt +; alt_tilt -; sufficiency: insufficient
"thin world... should_ai_abstain=true"
```

Listing 6: Raw grounding / action (same model family).

```
grounding: ready=[]; cannot confirm tilts
action (2026-01-17): bullish
"Positive 5-period momentum in raw feed."
```

5.3 RQ2 (safety valve): Hard refuse when the view is thin

Legibility does not imply safe action. Table 9 and Figure 4: on the same 100 days, Compiled abstains on every thin day (1.0); Ungated soft-judges at mean 0.68 (Gemini 0.43); Raw over-trades (Table 10); Blind abstains 1.0 with no feed—safe but useless as a data end. The costly failure mode is the middle case: half-baked Raw context makes models act and lose. Mean Δ CE (Compiled–Raw) +1.07 is avoided loss under the valve, not an alpha claim.

Full-OOS replication. Table 11: on all 2000 OOS asset-days (every day thin under the 0.2 gate), Compiled thin-abstain remains 1.000 (Wilson [.998, 1]) for every vendor. Ungated soft rates fall to a four-vendor mean ≈ 0.56 ; Raw GPT/Gemini abstain at 0.000. Scale does not rescue soft judgment. Blind refuses everywhere with no feed

Table 10: Raw action mix (100 days): half-baked context drives directional mass.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

Table 11: Full-OOS thin abstention ($n=2000$ asset-days; all thin under gate 0.2).

Model	Thin C	Thin U	Abs. R
gpt-5.4-mini	1.000	0.666	0.000
deepseek-v4-flash	1.000	0.540	0.748
glm-5.2	1.000	0.679*	0.078*
gemini-3.5-flash-lite	1.000	0.359	0.000
Mean	1.000	≈ 0.56	—

*GLM coverage incomplete (Ungated $n=1402$; Raw $n=675$).**Table 12: Open slice (3/3 ready $\equiv$ WMI ≥ 0.05 ; $n \approx 1224$). Content lower bound, not Compiled-open.**

Model	Abs. U	CE U	Act-CE U	Abs. R	CE R
gpt-5.4-mini	0.58	+0.42	+0.32	0.00	-0.28
deepseek-v4-flash	0.42	+0.21	+0.45	0.75	+0.04
glm-5.2	0.58	+0.24	+0.11	0.09*	-0.87*
gemini-3.5-flash-lite	0.34	+0.31	+0.06	0.00	-0.25

*Raw GLM $n=398$; Ungated GLM $n=932$; others $n=1224$. Bootstrap mean CE $\approx +0.31$ ($B=5,999$ reps; one-sided $p \approx 0.21$).

on the paired 100-day sample (abstain 1.0 for every vendor): the failure mode that needs a data end is not “models never abstain,” but “thin fragments make them act.”

Inside Ungated soft acts. On the mixed 100-day sample, when Ungated *does* act, act-conditional CE is sharply negative for three of four vendors (DeepSeek -3.12, GLM -4.28, Gemini -1.67) while GPT is positive on its smaller act set. Soft disclosure without a hard flag is not a free lunch: the same readable world that RQ1 scores can still harm decisions when the valve is removed.

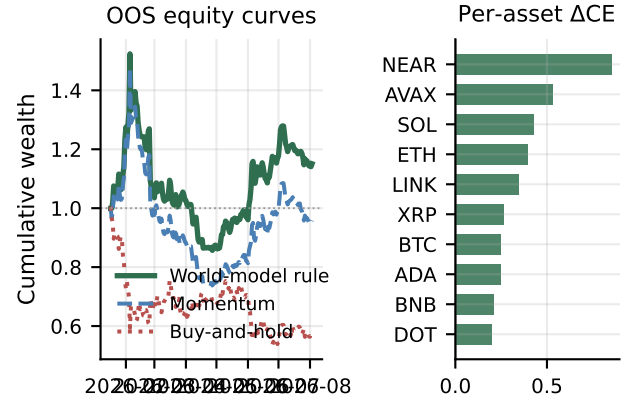
Open slice (content lower bound). Table 12: on WMI ≥ 0.05 ($\approx 61\%$ OOS; $n \approx 1224$) Ungated keeps compiled content without the hard boolean—mean abs. ≈ 0.48 , CE positive for every vendor (mean $\approx +0.29$; bootstrap $\approx +0.31$, CIs include 0). Production Compiled still abstains at 1.0; Raw stays risky (GPT/Gemini abs. 0.0, CE negative). Point-positive open-slice CE and negative mixed-sample act-conditional CE can coexist: regime and act selection matter; neither is a Compiled-open claim.

5.4 RQ3 (secondary): Nonempty bundle fields

A transparent no-LLM rule on vintaged macro_tilt / alt_tilt beats momentum on shared returns (Table 13, Figure 5): OOS Sharpe 0.764 / CE +0.130 vs. momentum (0.096, -0.206); block-bootstrap Δ CE=0.334 ($p=0.034$). A 2017–2026 audit with non-vintaged proxy

Table 13: No-LLM content probe (secondary). Bootstrap Δ CE 0.334; raw gap 0.336.

Policy	Sharpe	CE
Buy-and-hold	-1.404	-1.163
Momentum	0.096	-0.206
World-bundle tilts	0.764	+0.130
Tilts – Momentum	—	0.334 ($p=0.034$)

**Figure 5: Secondary content probe: compiled tilts beat momentum; Δ CE positive for every asset.****Table 14: Content probe by archive density (OOS).**

Slice	Share	Mech CE	Δ CE vs mom
All OOS	1.00	0.130	0.334*
Open / thick	0.61	0.061	0.315
Thin (exchange gap)	0.39	1.048	0.129

*Block-bootstrap Δ CE; raw mean gap 0.336.**Table 15: LOBO decomposition (ungated content probe). Content share 1.0.**

Drop	Δ CE content	p	Content share
macro	-0.334	0.034	1.0
alternative	-0.334	0.034	1.0
exchange	0.000	1.000	—

tilts shows no edge: value appears where the vintage-compiled world exists.

Density split and LOBO channel. Table 14: on the open/thick slice Δ CE=0.315; on exchange-gap days the rule abstains more (Δ CE=0.129)—complementary to the valve test. Table 15 and Figure 6 (Prop. 3.7): deleting macro or alternative collapses the gap through the *content* channel (content share 1.0; gating residual 0)—fields are nonempty, and the refuse flag is not what creates the edge.

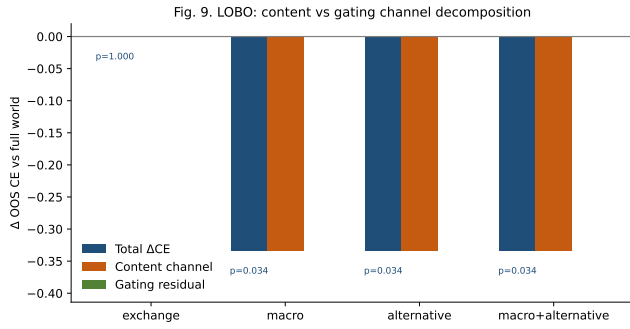


Figure 6: LOBO: deleting macro/alternative removes the edge via content, not the refuse flag.

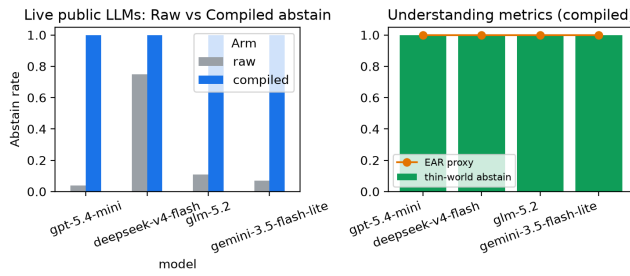


Figure 7: Seeing ladder (offline protocol check): compiled world fields are readable; raw momentum wiring is not a substitute world.

5.5 Discussion

What carries the paper. (i) The data end is legible: checkable grounding at ≈ 0.97 , with sufficiency chat shown to be a poor substitute (RQ1). (ii) Thin views must not drive action: soft text fails at 100-day and full-OOS scale; Blind shows models can refuse with no feed, but Raw shows thin fragments make them act and lose; the hard valve works (RQ2). (iii) Fields are nonempty under a no-LLM probe, density split, and LOBO content share 1.0; open-slice Ungated is a lower bound, not Compiled-open (RQ3). Relative to FinMem /FinAgent and ICAIF agent stacks [1, 3, 15, 16, 20, 21], we supply the observation stage those systems assume—not a competing decision or execution layer.

Why grounding first. Trading PnL confounds observation quality with policy skill. Grounding asks whether the consumer can audit ready/missing/signed state; Raw cannot, Compiled can. The valve then asks whether soft judgment stops action on a thin disclosed world; empirically, no—at either sample size. That ordering matches the stack: a decision engine that cannot see, or that acts on thin support, is not rescued by better execution.

Design. World bundle not feature dump; previous-close PIT with explicit freshness; four arms separate seeing+gating, seeing without gating, thin wiring, and no feed. Schema honesty: eight bands in the runtime, three in the evaluation archive; WMI discloses the resulting ceiling. Service-layer health and bundle reads make the

valve and the view callable by any downstream policy without re-ingesting collectors.

Reproducibility. Paper-lab smoke checks BandPIT/WMI/ACWMI wiring; PIT rebuild and the frozen four-arm harness regenerate the panel and live transcripts used here. Prompts, IS/OOS cut, seeds, and vendor IDs are fixed; credentials stay in environment variables. An anonymized harness ships on acceptance.

Scope. Estimand is data-end legibility and valve reliability on a scarce three-band archive—not decision-layer SOTA, not execution, and not production Compiled-open alpha. Denser historical bands would lift WMI; decision and execution stacks can sit on top once the view opens.

6 Conclusion

End-to-end crypto quant for public LLMs needs a reliable data end before decision or execution. We contribute that observation layer: compile asynchronous evidence into a consumable PIT world bundle, disclose structural missingness, and hard-refuse when the view is too thin. Grounding recovers ready/missing/tilt at ≈ 0.97 under Compiled vs. Raw collapse; sufficiency chat alone fails; the hard valve abstains 1.0 on thin days (including full OOS) where soft judgment does not; vintaged tilts are nonempty ($\Delta CE = 0.334$; LOBO content share 1.0). Build the quant data end first: compile so models can see; refuse when they must not act.

References

- [1] Chanyeol Choi, Jihoon Kwon, Alejandro Lopez-Lira, Chaewoon Kim, Minjae Kim, Juneha Hwang, Jaeseon Ha, Hojun Choi, Suyeol Yun, Yongjin Kim, and Yongjae Lee. 2025. FinAgentBench: A Benchmark Dataset for Agentic Retrieval in Financial Question Answering. In *Proceedings of the 6th ACM International Conference on AI in Finance*. ACM, 632–637. doi:10.1145/3768292.3770362
- [2] Dean Croushore and Tom Stark. 2001. A Real-Time Data Set for Macroeconomists. *Journal of Econometrics* 105, 1 (2001), 111–130.
- [3] Yitong Duan, Chuheng Zhang, and Jian Li. 2025. FactorMAD: A Multi-Agent Debate Framework Based on Large Language Models for Interpretable Stock Alpha Factor Mining. In *Proceedings of the 6th ACM International Conference on AI in Finance*. ACM, 605–613.
- [4] Ran El-Yaniv and Yair Wiener. 2010. On the Foundations of Noise-Free Selective Classification. *Journal of Machine Learning Research* 11 (2010), 1605–1641.
- [5] Yonatan Geifman and Ran El-Yaniv. 2017. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems*, Vol. 30. Curran Associates.
- [6] David Ha and Jürgen Schmidhuber. 2018. Recurrent World Models Facilitate Policy Evolution. In *Advances in Neural Information Processing Systems*, Vol. 31. Curran Associates.
- [7] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. 2020. Dream to Control: Learning Behaviors by Latent Imagination. In *International Conference on Learning Representations*. OpenReview.net.
- [8] Campbell R. Harvey, Yan Liu, and Heqing Zhu. 2016. ... and the Cross-Section of Expected Returns. *Review of Financial Studies* 29, 1 (2016), 5–68.
- [9] Yann LeCun. 2022. A Path Towards Autonomous Machine Intelligence. *OpenReview preprint* (2022).
- [10] Yang Li, Yangyang Yu, Haohang Li, Zhiyang Chen, and Khaldoun Khashanah. 2023. TradingGPT: Multi-Agent System with Layered Memory and Distinct Characters for Enhanced Financial Trading Performance. *arXiv preprint arXiv:2309.03736* (2023).
- [11] Yukun Liu, Aleh Tsyvinski, and Xi Wu. 2022. Common Risk Factors in Cryptocurrency. *Journal of Finance* 77, 2 (2022), 1133–1177.
- [12] Alejandro Lopez-Lira and Yuehua Tang. 2023. Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models. *arXiv preprint arXiv:2304.07619* (2023).
- [13] Igor Makarov and Antoinette Schoar. 2020. Trading and Arbitrage in Cryptocurrency Markets. *Journal of Financial Economics* 135, 2 (2020), 293–319.
- [14] Dimitris N. Politis and Joseph P. Romano. 1994. The Stationary Bootstrap. *J. Amer. Statist. Assoc.* 89, 428 (1994), 1303–1313.

- [15] Yiqing Shen, Jingshu Zhang, Feng Chen, Kaiyuan Yan, and Hongguang Li. 2025. FinSearch: A Temporal-Aware Search Agent Framework for Real-Time Financial Information Retrieval with Large Language Models. In *Proceedings of the 6th ACM International Conference on AI in Finance*. ACM. doi:10.1145/3768292.3770382
- [16] Rui Sun, Zuo Bai, Wentao Zhang, Yuxiang Zhang, Li Zhao, Shan Sun, and Zhengwen Qiu. 2025. FinResearchBench: A Logic Tree based Agent-as-a-Judge Evaluation Framework for Financial Research Agents. In *Proceedings of the 6th ACM International Conference on AI in Finance*. ACM, 656–664. doi:10.1145/3768292.3770364
- [17] Halbert White. 2000. A Reality Check for Data Snooping. *Econometrica* 68, 5 (2000), 1097–1126.
- [18] Hongyang Yang, Xiao-Yang Liu, and Christina Dan Wang. 2023. FinGPT: Open-Source Financial Large Language Models. In *FinLLM Symposium at IJCAI*.
- [19] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations*. OpenReview.net.
- [20] Yangyang Yu, Haohang Li, Zhiyang Chen, Yuechen Jiang, Yang Li, Denghui Zhang, Rong Liu, Jordan W. Suchow, and Khaldoun Khashanah. 2024. FinMem: A Performance-Enhanced LLM Trading Agent with Layered Memory and Character Design. *Proceedings of the AAAI Symposium Series* 3, 1 (2024), 595–597.
- [21] Wentao Zhang, Lingxuan Zhao, Hao Xia, Shuo Sun, Jiaze Sun, Molei Qin, Xinyi Li, Yuqing Zhao, Yilei Zhao, Xinyu Cai, et al. 2024. A Multimodal Foundation Agent for Financial Trading: Tool-Augmented, Diversified, and Generalist. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. ACM.